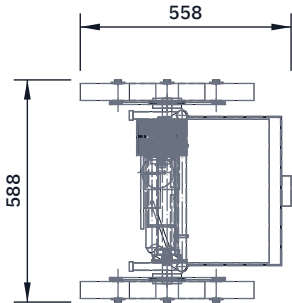


CONCEPT, NOT FOR FABRICATION

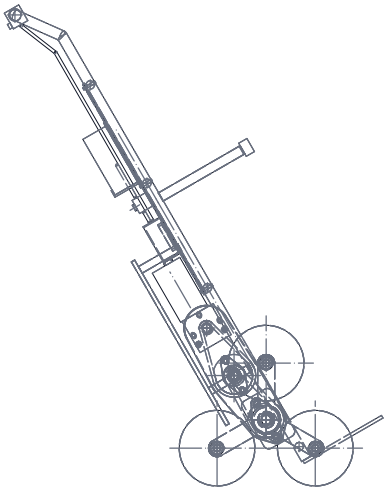
StepClimber · github.com/BoujeeEnjinia1701/stepclimber

REV	DESCRIPTION	DATE	BY
P1	Preliminary general arrangement, TRL 3	2026-09-25	AC
P2	200 mm wheels, 150 mm arms, two-stage drive (SCM-DDR-002)	2026-09-25	AC
P3	Layout and labels tidied	2026-09-30	AC
P4	Constructable design: axle plates, chain case, uprights (SCM-DDR-003)	2026-10-01	AC



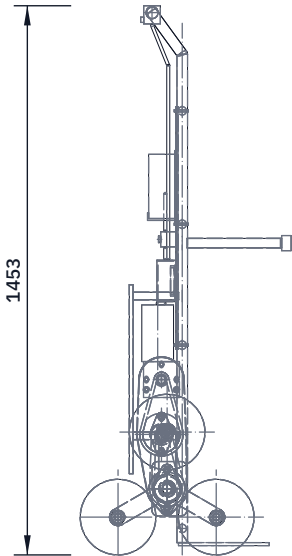
TOP VIEW

Scale 1:20; looking down (along -Z)



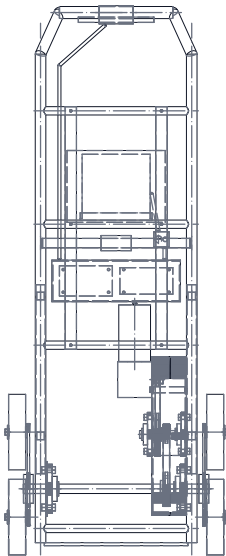
FRONT VIEW, CLIMBING POSE

Tilted 30 deg about the shaft; scale 1:20; looking at the front (along +Y)



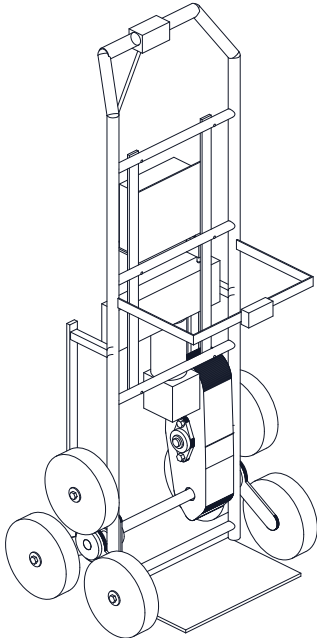
FRONT VIEW

Scale 1:20; looking at the front (along +Y)



RIGHT VIEW

Scale 1:20; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Overall width 569 (wheel outer faces); upright height 1451

Clusters: arm 150, wheel dia 200, spacing 259.8

Shaft 25 dia at 175.0 above floor; bearings 336 apart, inside the rails

Cluster mid-planes 524 apart; rails 400 apart, 28 tube

Drive 06B 10T:35T then 08B 10T:20T (7:1)

Countershaft 143.9 and gearmotor 289 up the frame, 15 back

Chains 38 links 08B, 54 links 06B; closed chain case

Case radius 53.9 on the shaft line (limit 54.2, R15 met)

Grip 1260 from the shaft along the frame; toe plate 240 x 380

Pack 75 x 180 x 170 on uprights, centre 800 above shaft

Width 588 over the wheel end screws; truck 34.2 kg (SCM-CAL-001 v0.3)

Climbing tilt 30 deg back; tilt window +/-6 deg; 60 kg rated stair load

STEPCLIMBER

github.com/BoujeeEnjinia1701/stepclimber

General arrangement

DRAWING NO. SCM-DWG-001			REV P4	SHEET 1 of 1	SIZE ANSI B
SCALE 1:20	UNITS mm	PROJECTION 	DATE 2026-10-01		DRAWN Amish Chadha
MATERIAL / NOTES Steel frame and spiders; solid rubber wheels; see bom/bom.csv. Figures from SCM-CAL-001 v0.3					
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